

Retail Sales Analytics and Customer Purchasing Behaviour Analysis

CIND820 – Big Data Analytics Project

Milestone 2: Architecture & Data Audit

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1. Introduction and Project Update

This interim report presents the analytical architecture and data-audit plan for the CIND820 final project, **Retail Sales Analytics and Customer Purchasing Behaviour Analysis**. The project is developed from the perspective of a retail data analyst supporting inventory planning, merchandising, marketing, sales-performance, and customer-retention decisions for an online retailer.

The analysis uses the Online Retail dataset, which contains transaction-level records from a United Kingdom-based online retailer between December 2010 and December 2011. The dataset supports analysis of product revenue, geographic market contribution, monthly sales patterns, transaction characteristics, and customer purchasing behaviour.

The business problem is that retail managers require reliable evidence to determine:

- which products should receive inventory and merchandising priority;
- which geographic markets contribute the greatest value;
- when seasonal demand is strongest;
- and which customer groups should receive targeted retention or marketing attention.

A basic descriptive analysis can identify products, countries, and customers with the highest historical revenue. However, revenue ranking alone does not explain how recently customers purchased, how frequently they purchased, or whether their spending behaviour is consistent. The planned approach therefore combines exploratory data analysis, data-quality auditing, revenue and trend analysis, and RFM customer segmentation.

Initial profiling identified missing customer identifiers, duplicate rows, cancelled invoices, negative quantities, zero or negative unit prices, incomplete product descriptions, administrative product labels, extreme transaction values, and an incomplete final month. These conditions affect the reliability of revenue calculations and customer-level conclusions and therefore require documented cleaning and analytical controls.

The purpose of Milestone 2 is to demonstrate that the proposed methodology is technically appropriate, transparent, reproducible, and connected to specific retail decisions.

2. Milestone 1 Feedback and Project Revisions

The project was revised in response to the Milestone 1 feedback.

The professional context was made more specific by defining the project from the perspective of a retail data analyst rather than describing a general academic analysis. The primary stakeholders are inventory and merchandising managers, marketing managers, customer-retention teams, and sales-performance managers.

The research questions were revised to connect each analytical output to a business decision. Product analysis is linked to inventory prioritization, country analysis to market concentration and diversification, monthly analysis to seasonal planning, and customer segmentation to retention and marketing decisions.

The dataset biography was expanded using actual profiling evidence. The raw dataset contains 541,909 transaction records and eight original variables. The audit identified 135,080 missing CustomerID values, 1,454 missing product descriptions, 5,268 duplicate records, 10,624 negative-quantity records, 9,288 cancelled invoice records, and 2,517 records with zero or negative unit prices.

The ethics and governance discussion were strengthened by recognizing CustomerID as an indirect identifier. Customer-level findings will therefore be reported as aggregate segments rather than as identifiable individual profiles.

A GitHub repository was also created to support reproducibility. It contains the raw dataset, original Milestone 1 report, initial EDA notebook, data-cleaning notebook, and business EDA notebook.

3. Professional Context and Decision Stakeholders

The project supports four primary stakeholder groups.

Inventory and merchandising managers need evidence about high-revenue products and seasonal changes so that stock availability and merchandising attention can be prioritized.

Marketing managers need evidence about geographic sales concentration and customer value so that campaigns can be directed toward relevant markets and segments.

Customer-retention teams need a transparent way to distinguish recent, frequent, and high-value customers from inactive or low-engagement customers.

Sales-performance managers need a reproducible view of revenue patterns and data-quality limitations before using the results in planning or performance discussions.

The analyst's role is to transform transaction-level data into decision-ready evidence while clearly separating observed patterns from claims that the dataset cannot support.

4. Updated Research Questions

1. Which merchandise products generate the highest revenue and should receive priority in inventory and merchandising decisions?
2. Which countries contribute the highest share of revenue, and what does this concentration imply for market dependence and diversification?
3. What monthly sales patterns can support seasonal demand and inventory planning?
4. How can customers be grouped using recency, frequency, and monetary (RFM) value to support retention and targeted marketing decisions?
5. Which data-quality and coverage limitations affect the reliability and generalizability of the findings?

5. Methodological Rationale & Literature Review

The methodological plan combines descriptive sales analysis with RFM customer segmentation.

The descriptive component provides a transparent baseline by ranking products and countries according to total revenue and examining monthly trends. This method is easy to interpret and appropriate for identifying historical performance, but it does not distinguish between customers who purchased recently, customers who purchase repeatedly, and customers whose total spending is high.

RFM analysis addresses this gap by measuring three customer-level dimensions:

- **Recency:** the time since the customer's most recent purchase;
- **Frequency:** the number of distinct purchasing occasions;
- **Monetary value:** the customer's total valid spending.

Prior retail analytics research has used RFM to quantify customer value from transaction data and to support customer-oriented marketing decisions. Studies using the same UK Online Retail dataset have also demonstrated that RFM variables can provide a practical foundation for customer segmentation and comparison of segmentation methods.

RFM is selected because it matches the available dataset. The data contains customer identifiers, invoice numbers, dates, quantities, and prices, but does not contain customer demographics, acquisition channels, satisfaction scores, or product cost information. RFM therefore uses variables that can be calculated directly and transparently from the available transactions.

The method is also consistent with the project's "no black box" requirement. Each score can be traced to observable purchasing behaviour and explained to a business stakeholder. More advanced clustering methods can capture complex group structures, but their results depend on scaling choices, the selected number of clusters, and algorithm assumptions. Research comparing clustering approaches on the UK retail dataset shows that different algorithms can produce different segmentation quality, which reinforces the need for a transparent baseline before considering more complex alternatives.

Exploratory data analysis is performed before segmentation because invalid records, missing identifiers, cancellations, and extreme values can distort revenue totals and customer metrics. Retail case evidence also shows that EDA and basic statistical analysis can support operational decisions such as ordering and inventory improvement.

5.1 Descriptive Revenue Analysis as the Baseline

The baseline method for this project is descriptive revenue analysis. Valid transactions are aggregated by product, country, month, and customer to identify the largest historical revenue contributors. This method is transparent, reproducible, and easy for retail decision-makers to understand.

However, revenue ranking provides only a partial view of customer value. It does not distinguish between a customer who purchased recently, a customer who purchases repeatedly, and a customer who made one large purchase in the past. Therefore, revenue alone is suitable as a reporting baseline but is not sufficient for customer-retention or targeted-marketing decisions.

5.2 RFM as the Selected Customer-Analysis Method

RFM analysis represents customer behaviour using recency, frequency, and monetary value. Recency measures how recently a customer made a purchase, frequency measures how often the customer purchased, and monetary value measures the customer's total spending during the observation period.

Christy et al. (2021) applied RFM analysis to transactional data and showed that these three behavioural measures can be used to support customer segmentation. This is relevant to the current project because the Online Retail dataset contains CustomerID, InvoiceNo, InvoiceDate, Quantity, and UnitPrice, which are sufficient to calculate the required RFM variables.

Similarly, Chen, Sain, and Guo (2012) applied RFM variables with K-means clustering in an online retail setting and demonstrated how transaction data can be converted into meaningful customer segments for customer-focused marketing.

RFM is a suitable method for this dataset because it relies on observable purchasing behaviour rather than demographic information that is not available. Compared with revenue-only ranking, RFM provides greater decision value by distinguishing customers according to how recently they purchased, how frequently they purchased, and how much they spent.

However, RFM describes observed transaction behaviour only. It does not explain why customers behave differently and should not be interpreted as a complete measure of customer lifetime value.

5.3 Comparison with K-Means Clustering-Based Segmentation

RFM variables are often combined with clustering algorithms to create customer groups with similar purchasing behaviour. Christy et al. (2021) Figure1. first applied RFM analysis to transactional data and then compared traditional K-means with Fuzzy C-means clustering. The study evaluated the methods using execution time, number of iterations, and cluster compactness.

This comparison is relevant to the current project because K-means could identify customer groups that are not visible through fixed RFM scoring rules. However, clustering introduces additional decisions, including feature scaling, selection of the number of clusters, centroid initialization, and validation of cluster quality.

For this project, transparent RFM scoring is selected as the primary customer-segmentation method because it is easier to interpret, reproduce, and explain to retail stakeholders. K-means is considered a more complex alternative that may provide additional insight, but it will not be assumed to be superior unless its clusters are validated and shown to offer greater business

value. This approach supports the project's no-black-box requirement and ensures that the final segment definitions remain understandable and defensible.

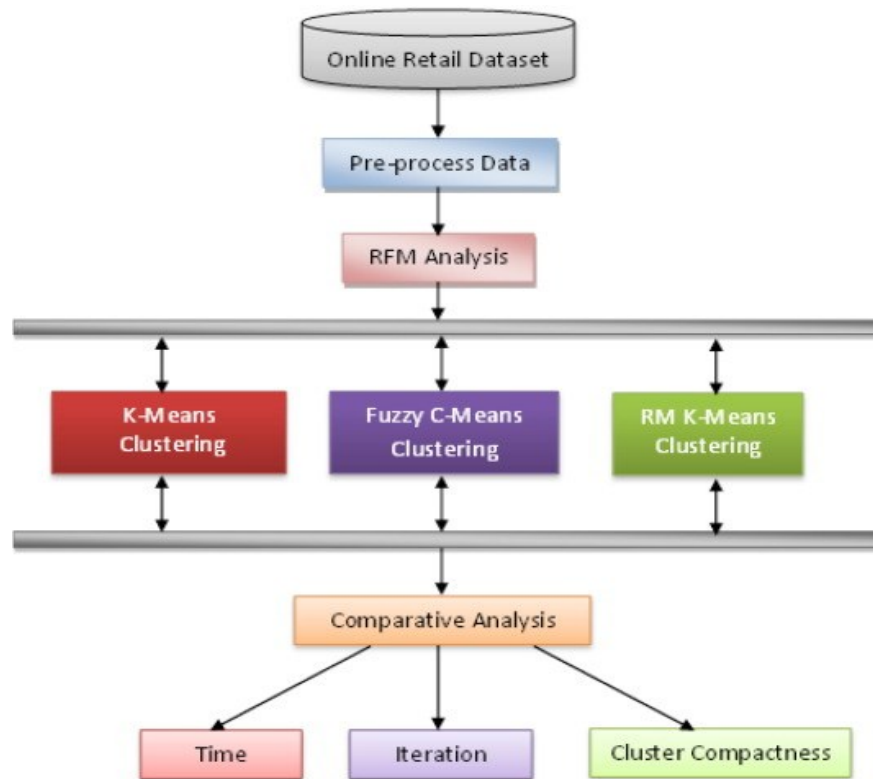


Figure1. Online Retail Dataset Analysis

5.4 Limitations of Standard RFM

Yang (2004) notes that traditional RFM is simple and actionable, but its scoring rules may be subjective and may produce overlapping or statistically weak segments. This limitation is relevant to the current project because fixed RFM thresholds may affect segment sizes and interpretation. Therefore, the final analysis will review segment distributions and test whether the selected scoring thresholds produce meaningful and stable customer groups.

Standard RFM also does not capture every aspect of customer behaviour. Alves Gomes and Meisen (2023) explain that customer segmentation may incorporate additional behavioural, demographic, interaction-based, and contextual variables. Extended transaction-based models may also include relationship length, purchase periodicity, product preferences, and returns behaviour.

These limitations are relevant to the current dataset. The available records cover approximately one year, December 2011 is incomplete, and many transactions have missing CustomerID values. Therefore, the project will not claim that RFM measures complete customer lifetime value. It will be used as a transparent behavioural segmentation method for the observed period only.

5.5 Data Profiling and Preparation Rationale

Data preparation is required before analysis because raw data may contain missing values, inconsistencies, invalid records, and structural issues that can affect analytical results. Fernandes et al. (2023) define data preparation as the process of transforming data from its existing form into a representation that is suitable for analysis. Their review identifies data profiling, format transformation, mapping, and data repair as central components of a preparation pipeline.

Clemente et al. (2023) further explain that systematic profiling supports data-centric analysis by identifying missingness, unusual distributions, structural problems, and other quality issues before later analytical stages are performed.

This rationale directly applies to the Online Retail dataset. Initial profiling identified duplicate rows, missing customer identifiers, cancelled invoices, negative quantities, zero or negative prices, administrative product descriptions, and extreme transaction values. These issues could distort revenue calculations, product rankings, and customer segmentation if they were not addressed before feature engineering.

The project therefore follows a structured preparation sequence: profile the raw data, validate variable types and business rules, remove or separate invalid transactions, create derived variables such as Revenue and Month, and build a separate customer-level dataset for RFM analysis. This sequence ensures that later analytical outputs are based on consistent and analysis-ready data.

5.6 Final Methodological Choice

The literature review supports a combined analytical approach based on descriptive revenue analysis and RFM customer segmentation. Descriptive analysis is retained as the baseline because it provides a simple, transparent, and reproducible view of product, country, monthly, and customer revenue performance.

RFM is selected as the primary customer-analysis method because it adds recency and frequency to monetary value, allowing customer behaviour to be assessed more comprehensively than through revenue ranking alone. It is also well matched to the available transaction-level variables and does not require demographic or marketing-response data that are absent from the dataset.

Clustering methods such as K-means may identify additional structure in the customer data, but they introduce further methodological choices related to scaling, cluster selection, initialization, and validation. Therefore, they are treated as more complex alternatives rather than automatically superior methods.

The final architecture is designed to balance business usefulness, methodological transparency, and reproducibility. It supports the project's no-black-box requirement and ensures that each analytical result can be traced back to documented data-preparation steps and observable transaction behaviour.

6. Baseline Method and Proposed Analytical Approach

6.1 Baseline Method: Descriptive Revenue Analysis

The baseline method is descriptive revenue analysis. Valid transactions will be aggregated by product, country, month, and customer to identify the highest revenue contributors and major sales patterns.

This method is appropriate as a baseline because it is transparent, reproducible, and easy for business stakeholders to interpret. It directly supports decisions related to product performance, geographic market contribution, and seasonal revenue trends.

However, revenue ranking alone provides an incomplete view of customer value. It does not distinguish between a customer who purchased recently, a customer who purchases frequently, and a customer who made one large purchase during the observation period.

6.2 Proposed Analytical Approach: Descriptive Analysis and RFM Segmentation

The proposed approach combines descriptive revenue analysis with RFM customer segmentation.

The descriptive component will examine:

- product revenue performance;
- country-level revenue concentration;
- monthly revenue trends;
- transaction distributions and variable relationships.

The RFM component will calculate:

- **Recency:** number of days since the customer's most recent valid purchase;
- **Frequency:** number of distinct valid invoices associated with the customer;
- **Monetary value:** total valid revenue generated by the customer.

Customers will then be grouped according to their RFM characteristics to identify segments such as high-value customers, loyal customers, recent customers, and customers at risk of becoming inactive.

6.3 Comparison of the Two Approaches

Evaluation area	Descriptive revenue baseline	Proposed RFM approach
Main focus	Historical revenue	Customer purchasing behaviour
Customer value measure	Total spending only	Recency, frequency, and spending
Interpretability	High	High
Data requirements	Basic transaction data	Customer ID, invoice, date, and revenue
Business use	Product, market, and trend reporting	Retention and targeted marketing
Main limitation	Does not capture recent or repeated activity	Does not explain why behaviour occurs

The proposed RFM approach improves on the baseline by adding customer recency and purchase frequency to monetary value. This provides more useful evidence for customer-retention and targeted-marketing decisions while maintaining transparency.

6.4 Role of K-Means as an Alternative

K-means clustering is recognized as a more complex alternative that could identify customer groups based on similarities in their RFM values. However, it requires feature scaling, selection of the number of clusters, centroid initialization, and cluster-quality validation.

For this project, RFM scoring is selected as the primary segmentation method because it is easier to reproduce and explain to retail stakeholders. K-means will not be treated as superior unless it is implemented, validated, and shown to provide additional business value.

7. Dataset Biography and Data Structure

7.1 Dataset Background

The project uses the **Online Retail dataset**, which contains transaction-level records from a United Kingdom-based online retailer. The data covers transactions from December 2010 to December 2011 and includes customers from multiple countries.

The dataset was originally collected to record retail sales transactions. Each row represents one invoice-line item rather than one complete order. Therefore, a single invoice may appear across several rows when multiple products were purchased in the same transaction.

The raw dataset contains **541,909 rows and eight original variables**.

The dataset was obtained from the UCI Machine Learning Repository(Chen, 2015) and is publicly available for research and educational use.

7.2 Variable Structure

The original variables are:

Variable	Description	Analytical Role
InvoiceNo	Unique invoice or transaction identifier	Used to identify purchases and calculate frequency
StockCode	Product code	Used to distinguish individual products
Description	Product description	Used for product-level analysis
Quantity	Number of units in the invoice line	Used in demand and revenue calculations
InvoiceDate	Date and time of the transaction	Used for monthly trends and recency
UnitPrice	Price per product unit	Used to calculate revenue
CustomerID	Customer identifier	Used for customer-level analysis and RFM segmentation
Country	Customer country	Used for geographic revenue analysis

A derived variable, Revenue, is calculated as:

Revenue = Quantity × Unit Price

Additional derived variables will include:

- Month, derived from Invoice Date;
- Recency, calculated from the customer's most recent valid purchase;
- Frequency, based on the number of distinct valid invoices;
- Monetary, based on total valid customer revenue.

7.3 Data Structure

The dataset is **structured tabular data** stored in Excel format. It does not contain JSON, text documents, images, or nested fields. Therefore, no semi-structured parsing or flattening is required.

The project uses one main data source, so no relational join or external data integration is required. However, aggregation is necessary because the raw data is stored at the invoice-line level.

The dataset will be transformed into several analytical views:

- product-level revenue table;
- country-level revenue table;
- monthly revenue table;
- customer-level RFM table.

7.4 Data Preparation Requirements

The raw dataset cannot be used directly because it contains records that may not represent completed retail sales.

The required preparation steps include:

1. removing duplicate rows;
2. separating or excluding cancelled invoices;
3. removing negative and zero quantity values from completed-sales analysis;
4. removing zero or negative unit prices;
5. reviewing missing product descriptions;

6. excluding administrative product descriptions where product rankings are required;
7. creating the Revenue variable;
8. converting InvoiceDate to a valid datetime format;
9. creating a separate customer-level dataset without missing CustomerID values.

Two cleaned datasets are used for different analytical purposes.

The **main cleaned sales dataset** retains valid sales records even when CustomerID is missing. These records remain useful for product, country, revenue, and monthly trend analysis.

The **customer-level dataset** excludes records with missing CustomerID values because customer segmentation cannot be completed reliably without a customer identifier.

7.5 Dataset Suitability

The dataset is suitable for the project because it contains the variables required for revenue analysis, time-based analysis, geographic analysis, and RFM segmentation.

It is particularly well suited to behavioural customer analysis because it includes customer identifiers, invoice numbers, transaction dates, quantities, and prices.

However, the dataset does not contain demographic information, product cost, profit margin, customer satisfaction, marketing-channel data, or campaign-response variables. Therefore, the project can describe transaction behaviour and revenue patterns but cannot explain the underlying reasons for customer behaviour or calculate profitability.

8.Data Profiling and Audit

8.1 Initial Data-Quality Profile

The raw Online Retail dataset contains 541,909 transaction-line records and eight original variables. Initial profiling identified missing values, duplicate rows, cancellations, returns, and invalid pricing records that required investigation before analysis.

Data-Quality Issue	Records Identified	Analytical Impact
Missing CustomerID	135,080	Limits customer-level analysis and RFM segmentation
Missing Description	1,454	Limits product identification and product-level reporting
Duplicate rows	5,268	May overstate revenue and transaction volume
Negative Quantity	10,624	Likely represents returns, cancellations, or adjustments
Cancelled invoices	9,288	Do not represent completed sales
Zero or negative UnitPrice	2,517	May create invalid or misleading revenue values

These findings confirmed that the raw dataset could not be used directly. Duplicate records could result in double-counting, while cancelled invoices, negative quantities, and invalid prices could distort revenue and product-demand results. Missing customer identifiers also reduce the number of transactions that can support customer-level analysis.

8.2 Business-Rule Validation

The audit applied transaction-level business rules to distinguish completed sales from returns, cancellations, and administrative adjustments.

Invoices beginning with the letter C were treated as cancelled transactions. Records with negative quantities were interpreted as possible returns or reversals, while zero or negative unit prices were treated as invalid for completed-sales revenue analysis.

Administrative descriptions such as POSTAGE, DOTCOM POSTAGE, and Manual were also excluded from merchandise rankings because they do not represent normal retail products. This validation prevented non-product charges from being incorrectly reported as top-performing merchandise.

8.3 Univariate Analysis

The revenue distribution was strongly right skewed. Most transaction lines generated relatively low revenue, while a small number produced much larger values. The 99th percentile of transaction-line revenue was 183.60, meaning that approximately 99% of valid transaction-line revenue values were at or below this amount.

This result indicates that the mean revenue may be influenced by extreme transactions. Therefore, distributions, percentiles, and visual inspection should be considered alongside average values.

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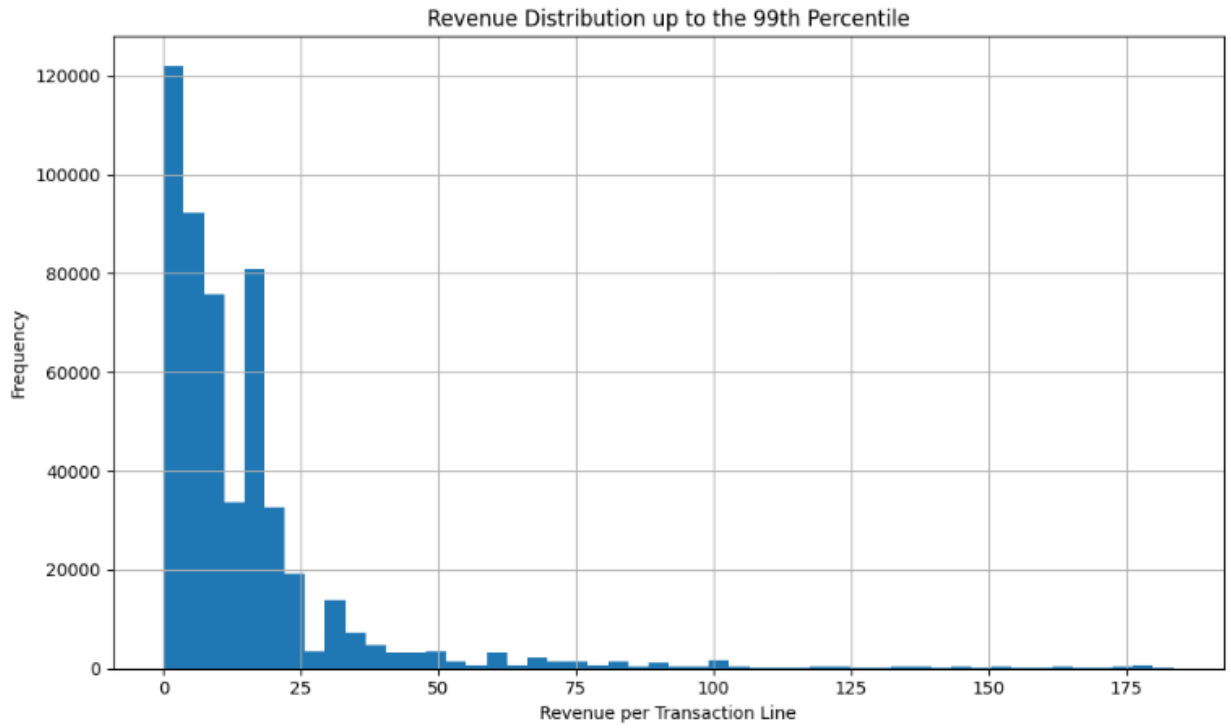


Figure2: Revenue Distribution up to the 99th Percentile

Monthly revenue also varied across the observation period. Revenue increased strongly from August to November 2011, with November recording the highest observed monthly revenue. The decline shown in December should not be interpreted as a full-month reduction because the dataset ends during December 2011.

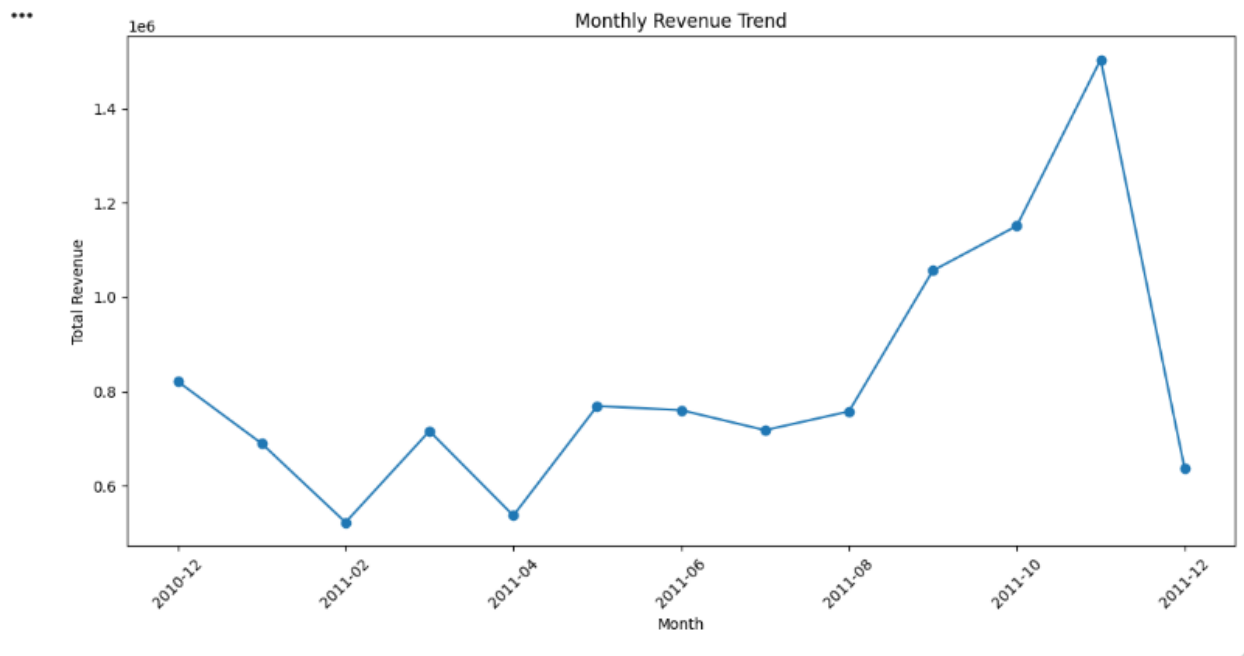


Figure3: Monthly Revenue Trend

8.4 Product and Geographic Analysis

Country-level profiling showed that revenue was highly concentrated in the United Kingdom. The Netherlands, EIRE, Germany, France, Australia, Spain, Switzerland, Belgium, and Sweden were among the next-highest contributors, but their revenue was substantially lower.

This concentration suggests that the retailer is strongly dependent on its domestic market. The finding may support continued attention to the United Kingdom while also identifying international markets for further investigation. However, the dataset represents one UK retailer, so the result cannot be generalized to the broader global retail sector.

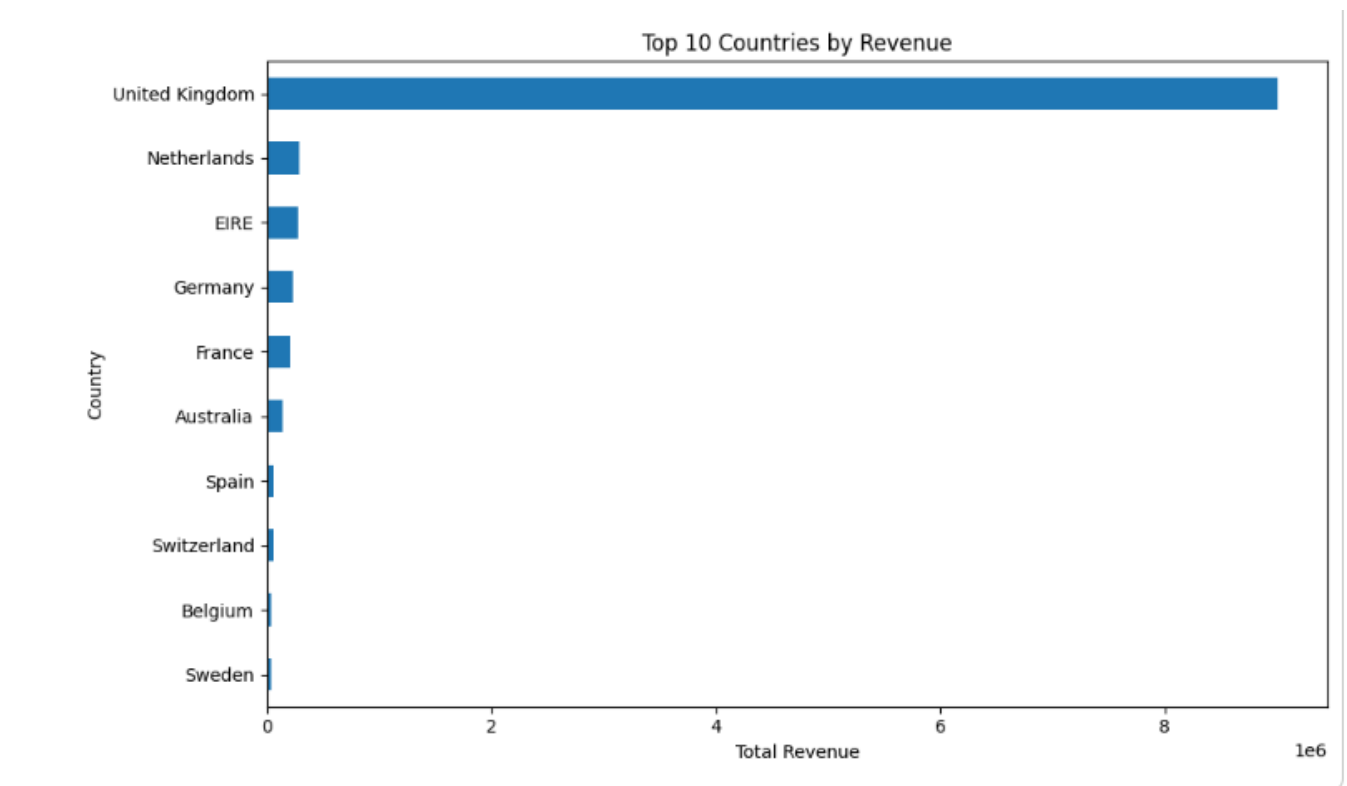


Figure 4: Top 10 Countries by Revenue

After excluding administrative descriptions, REGENCY CAKESTAND 3 TIER generated the highest merchandise revenue. Other leading products included PAPER CRAFT, LITTLE BIRDIE, WHITE HANGING HEART T-LIGHT HOLDER, PARTY BUNTING, and JUMBO BAG RED RETROSPOT.

These products may be considered for inventory and merchandising attention. However, total revenue alone does not indicate whether performance results from high quantity, high unit price, or both.

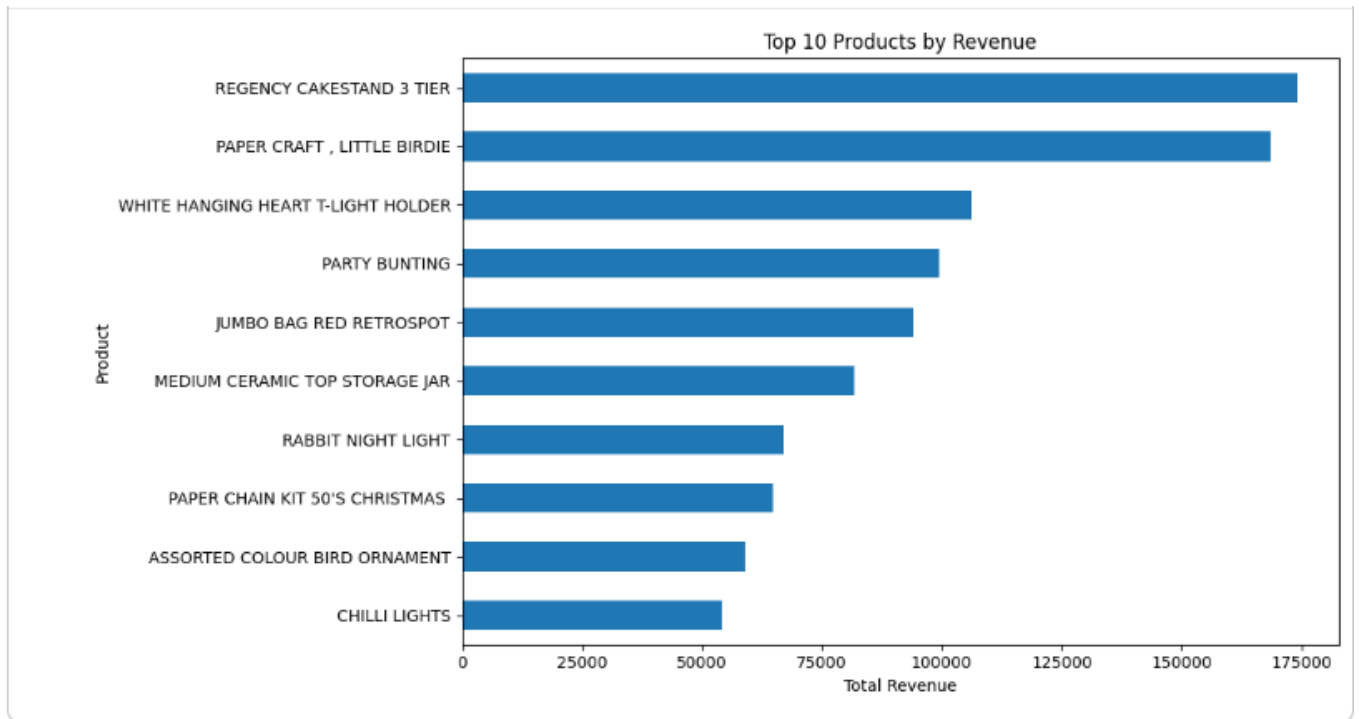


Figure 5: Top 10 Products by Revenue

8.5 Bivariate Analysis

The relationship between Quantity and Revenue was strongly positive, with a correlation of approximately 0.907. Transactions containing larger quantities generally generated higher revenue.

This result is expected because revenue is calculated using quantity and unit price. However, quantity should not be interpreted as the only driver of revenue because different products have different prices, and extreme transactions affect the observed pattern.

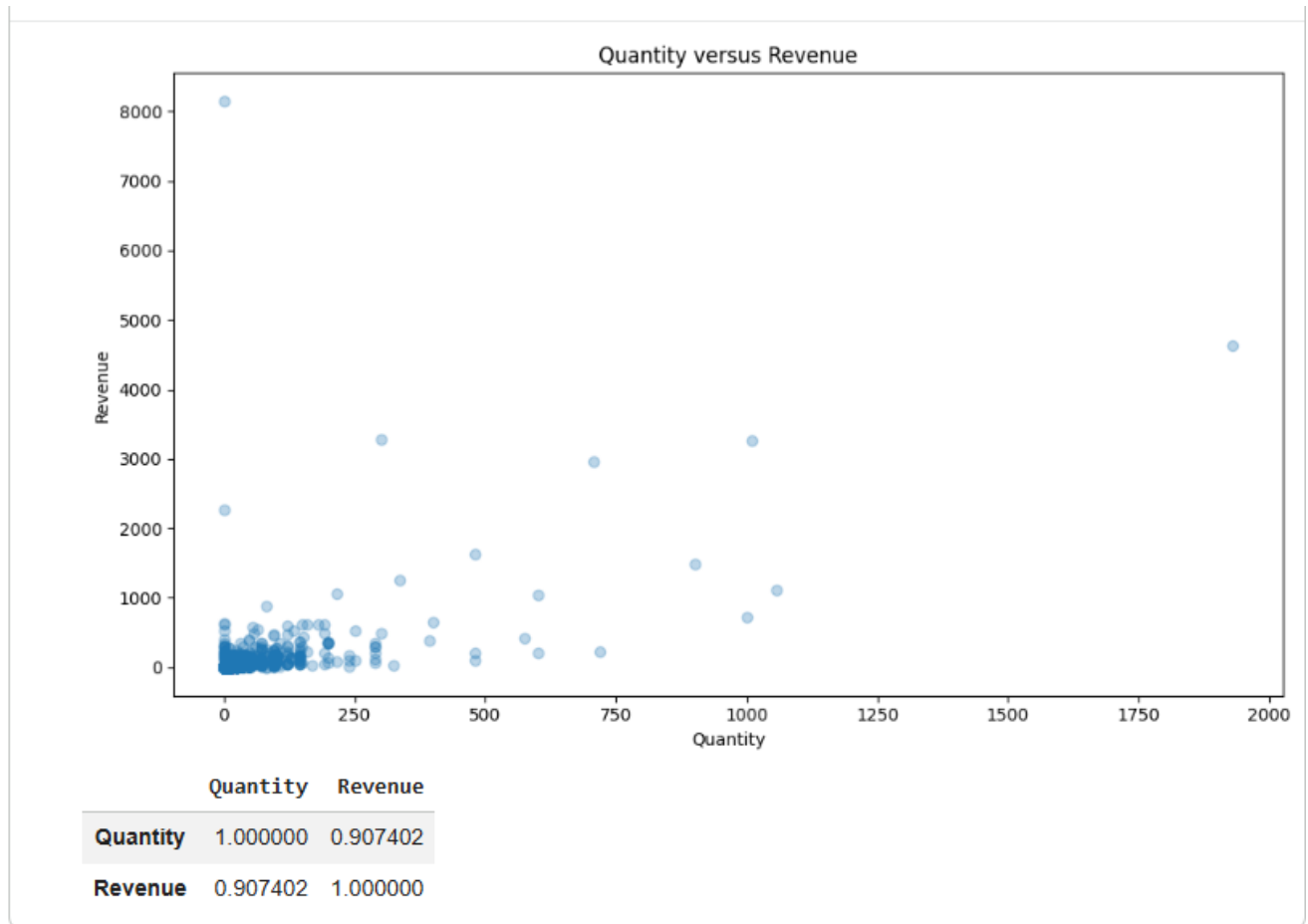


Figure 6: Quantity versus Revenue

The correlation between UnitPrice and Quantity was approximately -0.004 , indicating almost no overall linear relationship. Although some higher-priced products appeared to be purchased in smaller quantities, the relationship was too weak to support a general conclusion.

This suggests that purchasing quantity is influenced by additional factors such as product type, customer requirements, and order characteristics.

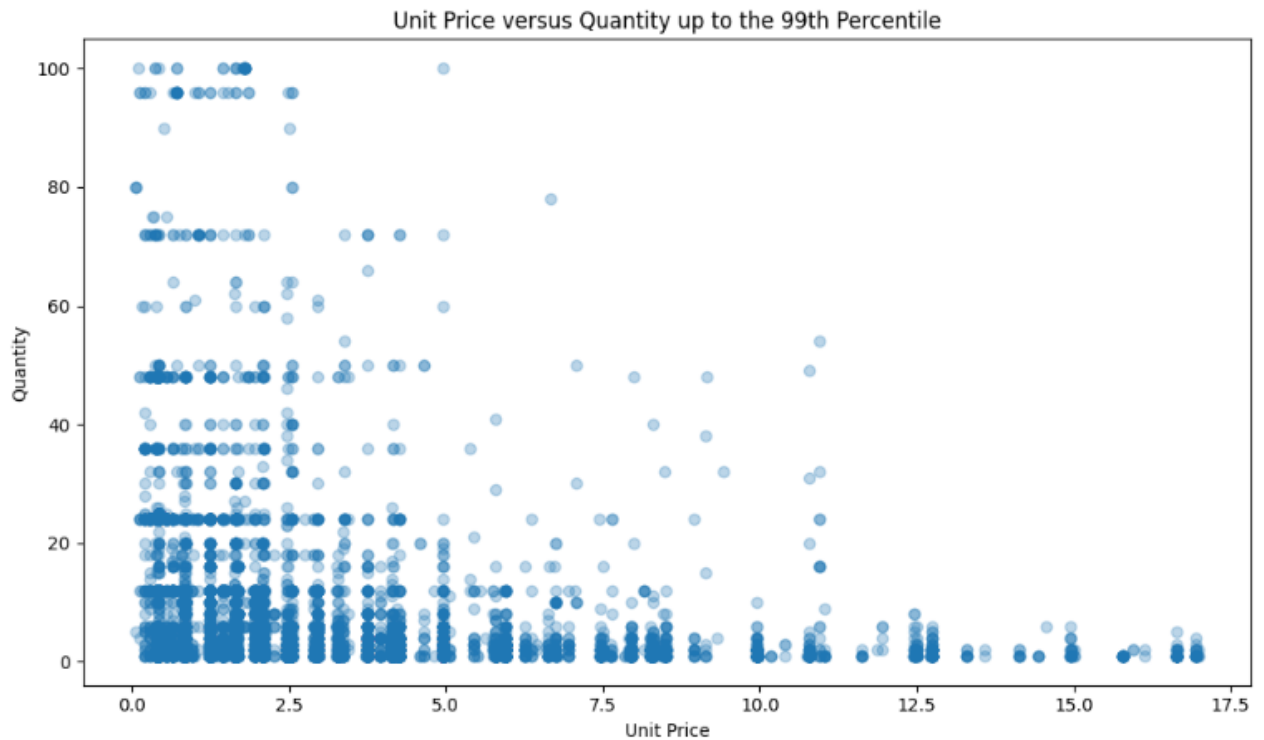


Figure 7: Unit Price versus Quantity

8.6 Results of Data Preparation

After removing duplicate rows, cancelled invoices, negative quantities, and zero or negative unit prices, the main cleaned sales dataset contained 524,878 records and nine variables, including the derived Revenue field.

A separate customer-level dataset was created by excluding records with missing CustomerID values. This dataset contains 392,692 transaction records and will be used for RFM analysis.

The two-dataset strategy preserves valid anonymous sales records for product, country, and trend analysis while ensuring that customer segmentation uses only transactions linked to a valid customer identifier.

8.7 Constraints Identified During Profiling

The profiling process identified several constraints that affect the conclusions:

- The dataset represents only one UK-based retailer.

- It covers approximately one year, so multi-year seasonal patterns cannot be confirmed.
- December 2011 is incomplete and cannot be compared directly with full months.
- Missing CustomerID values reduce the customer population available for segmentation.
- No demographic, satisfaction, marketing-channel, product-cost, or profit data is available.
- Returns and cancellations are excluded from the main completed-sales analysis.
- Revenue findings cannot be interpreted as profitability.

Therefore, the project can describe observed transaction behaviour and customer purchasing patterns, but it cannot explain the causes of customer behaviour or generalize results to all retailers.

9. Data Cleaning and Preparation

9.1 Cleaning Objective

The purpose of the cleaning process was to create reliable datasets for two different analytical needs:

1. general sales analysis, including product, country, monthly trend, and revenue analysis; and
2. customer-level analysis, including RFM segmentation.

The raw dataset could not be used directly because it contained duplicate rows, cancelled invoices, negative quantities, zero or negative prices, missing customer identifiers, and administrative product descriptions.

9.2 Cleaning Steps

The following preparation steps were applied in sequence:

Cleaning Step	Action Taken	Reason
Duplicate removal	Exact duplicate rows were removed	Prevents double-counting of transactions, volume, and revenue
Cancelled invoice filtering	Invoice numbers beginning with C were excluded from completed-sales analysis	These records represent cancellations or return rather than completed sales
Quantity validation	Records with Quantity less than or equal to zero were removed	Negative or zero quantities do not represent valid completed sales
Unit price validation	Records with Unit Price less than or equal to zero were removed	Prevents invalid or misleading revenue calculations
Revenue recalculation	Revenue = Quantity × Unit Price was calculated after filtering	Ensures revenue is based only on valid sales records
Date conversion	invoice Date was converted to datetime format	Supports monthly trend analysis and recency calculation
Administrative description filtering	Entries such as POSTAGE, DOTCOM POSTAGE, and Manual were excluded from merchandise rankings	Prevents service charges and adjustments from being treated as products

Customer identifier handling	Missing CustomerID values were excluded only from customer-level analysis	Preserves useful anonymous sales data while protecting the reliability of RFM analysis
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9.3 Creation of Two Analysis-Ready Datasets

Two cleaned datasets were created because the project includes both sales-level and customer-level analysis.

Main Cleaned Sales Dataset

The main cleaned sales dataset contains **524,878 records and nine variables**. It retains valid completed-sales records even when CustomerID is missing.

This dataset is used for:

- product revenue analysis;
- country-level analysis;
- monthly revenue trends;
- revenue distributions;
- transaction-level relationship analysis.

Records with missing CustomerID remain useful for these analyses because product, country, date, quantity, price, and revenue information are still available.

Customer-Level Dataset

A separate customer-level dataset was created by removing records with missing CustomerID values. This dataset contains **392,692 records and nine variables**.

This dataset will be used for:

- customer purchase aggregation;
- recency calculation;
- frequency calculation;
- monetary-value calculation;
- RFM segmentation.

A valid customer identifier is required because transactions cannot be grouped reliably at the customer level without it.

9.4 Feature Engineering

The following derived variables were created or planned:

Derived Variable	Calculation	Analytical Purpose
Revenue	Quantity × Unit Price	Measures transaction-line sales value
Month	Extracted from Invoice Date	Supports monthly trend analysis
Recency	Days since the customer's most recent valid purchase	Measures how recently the customer purchased
Frequency	Number of distinct valid invoices per customer	Measures repeat purchasing activity
Monetary	Total valid customer revenue	Measures customer spending value

Feature engineering occurs only after cleaning because invalid transactions could otherwise distort revenue totals and RFM values.

9.5 Handling Missing Values

Missing Description values were reviewed because they limit product identification. Such records are not suitable for product-name reporting unless the product can be identified through another valid field.

Missing CustomerID values were handled differently depending on the analytical purpose:

- retained in the main sales dataset;
- excluded from the customer-level dataset.

No artificial imputation was applied to CustomerID because assigning invented customer identifiers would create false customer histories and invalidate RFM segmentation.

9.6 Outliers and Extreme Values

The profiling stage identified strongly right-skewed revenue values and several extreme transactions.

These values were not automatically removed because some may represent legitimate high-volume purchases. Instead:

- the 99th percentile was used to improve visualization readability;
- extreme observations were reviewed through distributions and scatter plots;
- major business analyses were interpreted with caution.

This approach avoids deleting potentially valid commercial transactions solely because they are large.

9.7 Reproducibility

The full cleaning workflow is documented in the GitHub notebook:

`notebooks/Data_Cleaning_&_Preperation.ipynb`

The cleaned CSV files were not uploaded because they exceeded GitHub's web-upload limit. However, they can be reproduced by running the cleaning notebook using the raw Online Retail dataset.

This approach preserves reproducibility because another reviewer can follow the same steps and recreate both analysis-ready datasets.

9.8 Preparation Outcome

The cleaning and preparation strategy creates a clear separation between transaction-level sales analysis and customer-level behavioural analysis.

This design reduces unnecessary data loss, protects the validity of RFM calculations, and ensures that each analytical output is based on records appropriate for its intended purpose.

10. Data Constraints, Limitations & Privacy

10.1 Dataset Limitations

The Online Retail dataset is suitable for transaction-level sales analysis and customer-behaviour analysis, but several limitations affect the interpretation and generalizability of the results.

Constraint	Impact on Analysis	Impact on Claims
Data represents one UK-based retailer	Results reflect one specific business context	Findings cannot be generalized to all retailers or e-commerce businesses
Approximately one year of data	Limited historical coverage	Multi-year seasonality and long-term trends cannot be confirmed
December 2011 is incomplete	The final month contains partial data	December revenue cannot be compared directly with complete months
Missing CustomerID values	Reduces the number of transactions available for customer analysis	RFM findings apply only to customers with valid identifiers
Missing product descriptions	Some items cannot be identified clearly	Product-level conclusions exclude unidentified descriptions
No demographic variables	Age, gender, income, and household characteristics are unavailable	The analysis cannot explain why customers behave differently
No marketing-response data	Campaign exposure and response are unavailable	The project cannot measure campaign effectiveness
No product-cost or profit information	Revenue can be calculated, but margin cannot	High revenue cannot be interpreted as high profitability

Returns and cancellations excluded from main sales analysis	Main analysis focuses on completed positive sales	Results do not represent full net revenue or returns performance
One observation period	Customer history is limited to the available dates	RFM scores represent observed behaviour during this period, not complete lifetime value

These limitations mean that the project can describe observed sales patterns, market concentration, product revenue, and customer purchasing behaviour. It cannot establish causal relationships, explain customer motivations, calculate profitability, or claim that the findings represent the broader retail industry.

10.2 Privacy Assessment

The dataset does not contain direct personal identifiers such as names, addresses, telephone numbers, or email addresses. However, CustomerID functions as an indirect identifier because it links multiple transactions to the same customer.

The project will not attempt to identify individual customers or combine CustomerID with external information. Customer-level results will be reported only in aggregated form, such as RFM segments, segment sizes, average values, and revenue contribution.

Individual customer transaction histories will not be displayed in the final report or dashboard.

10.3 Data-Governance Controls

The following controls will be applied:

- the dataset will be used only for the stated analytical and educational purpose;
- no attempt will be made to re-identify customers;
- customer findings will be reported at segment level;
- only the minimum necessary variables will be used for each analysis;
- GitHub documentation will explain the analytical process without publishing individual customer profiles;
- derived outputs will avoid exposing unnecessary customer-level detail.

These controls are consistent with the principle that confidential information should be protected from unauthorized access, use, or disclosure.

10.4 Risk of Misuse

RFM segmentation may classify customers as high value, loyal, new, inactive, or at risk. These labels should be treated as analytical summaries rather than permanent characteristics.

The results should not be used to unfairly exclude customers from service, stereotype individuals, or make unsupported assumptions about their personal circumstances. The intended use is to support more relevant customer engagement, retention planning, and marketing prioritization.

10.5 Privacy and Limitation Conclusion

The dataset presents a low direct-identification risk because direct personal information is absent. However, **CustomerID** still requires careful handling as an indirect identifier.

The final analysis will therefore balance business usefulness with privacy protection by reporting aggregate customer segments and clearly stating the limits of the available data.

11. System Architecture and Analytical Pipeline

11.1 Analytical Pipeline Overview

Figure 8 presents the end-to-end analytical pipeline developed for the Online Retail project. The diagram shows how the raw transaction data progresses through data loading, profiling, privacy review, cleaning, feature engineering, descriptive analysis, RFM customer segmentation, evaluation, and final reporting.

The sequence is designed to ensure that data-quality issues are identified and corrected before analytical variables and customer segments are created. Each stage produces an output required by the next stage, supporting a transparent, reproducible, and business-focused analytical process.

The pipeline diagram was created in Lucid chart and is available through the project documentation. (["Lucid App Link"](#))

Figure 8: Pipeline Diagram

11.2 Sequence of Operations

The analytical workflow begins with the raw Online Retail dataset. The dataset is loaded into Python, and variable types are reviewed to confirm that transaction dates, identifiers, quantities, prices, and categorical fields are represented correctly.

Initial profiling is performed before any analysis. This stage identifies missing values, duplicate records, cancelled invoices, negative quantities, zero or negative prices, unusual distributions, and privacy concerns. Profiling must occur first because later revenue calculations and customer features depend on the quality of the underlying transaction records.

The next stage applies the documented cleaning rules. Exact duplicate rows are removed, cancelled invoices are excluded from completed-sales analysis, and records with invalid



quantities or prices are filtered. Revenue is recalculated only after these steps so that the resulting values reflect valid completed sales.

The cleaned data is then divided into two analytical views. The main cleaned sales dataset supports product, country, monthly, and transaction-level analysis. A separate customer-level dataset removes records with missing CustomerID values because customer aggregation and RFM segmentation require a valid identifier.

Feature engineering follows data cleaning. Revenue is calculated from quantity and unit price, Month is derived from invoice date, and the RFM variables are created from valid customer transactions. Recency depends on the latest valid purchase date, frequency depends on distinct valid invoices, and monetary value depends on total valid customer revenue.

The descriptive baseline analysis is completed before RFM segmentation. This provides a transparent comparison point by showing product, country, monthly, and customer revenue performance. RFM segmentation is then used to add recent activity and repeat purchasing behaviour to monetary value.

The final stages evaluate whether the resulting outputs are interpretable, reproducible, and useful for retail decisions. Findings are reviewed together with the identified limitations before they are included in the final report.

11.3 Dependencies Between Pipeline Steps

The steps in the pipeline are dependent on one another.

- Revenue cannot be calculated reliably until invalid quantities, prices, cancellations, and duplicates are addressed.
- Monthly analysis requires InvoiceDate to be converted to a valid datetime format.
- Customer-level analysis requires records with valid CustomerID values.
- Recency cannot be calculated until the observation date and each customer's latest valid purchase are defined.
- Frequency must be based on distinct valid invoices rather than transaction-line counts.
- Monetary value must be calculated from cleaned revenue rather than raw transaction values.
- RFM segmentation should occur only after the customer-level features have been reviewed for missingness, skewness, and extreme values.
- Business recommendations should be produced only after analytical results are interpreted alongside the dataset limitations.

This sequence prevents invalid source records from being carried forward into later analysis.

11.4 Leakage Prevention and Circular Reasoning

The current project is descriptive and unsupervised rather than predictive, so conventional target leakage is less relevant than it would be in a supervised machine-learning model. However, the workflow still includes controls to prevent circular or misleading analysis.

Revenue is not treated as an independent explanation of monetary value because monetary value is directly derived from revenue. Similarly, the strong relationship between quantity and revenue is interpreted cautiously because quantity is part of the revenue formula.

RFM features are created only from valid historical transactions. Segment labels are assigned after the RFM measures are calculated and are not used to generate those same measures. No future transactions outside the observation period are used to calculate customer behaviour.

If K-means clustering is implemented as an alternative, feature scaling will be fitted only to the RFM feature table used for clustering. The number of clusters will be selected using documented validation criteria rather than chosen solely to produce preferred business labels.

11.5 Training and Validation Strategy

A conventional training and testing split is not required for the descriptive baseline or rule-based RFM scoring because these methods do not predict a labelled outcome.

Instead, validation focuses on analytical quality and business interpretability.

The descriptive analysis will be validated by:

- confirming totals after data cleaning;
- comparing aggregated values with notebook outputs;
- reviewing whether administrative records distort product rankings;
- checking whether incomplete months affect trend interpretation.

RFM segmentation will be validated by:

- reviewing the distribution of Recency, Frequency, and Monetary values;
- checking the number and size of customer segments;
- comparing median RFM values across segments;
- examining revenue contribution by segment;
- testing whether segment labels match the underlying customer behaviour;
- assessing the sensitivity of segment definitions to score thresholds.

If K-means is implemented, additional validation will include feature scaling, elbow analysis, silhouette score, cluster stability, and business interpretability.

11.6 Evaluation Strategy

The baseline and proposed methods will be evaluated against their ability to answer the research questions.

Evaluation Area	Descriptive Baseline	RFM Approach
Product prioritization	Revenue ranking by product	Not the main purpose
Market performance	Revenue ranking by country	Not the main purpose
Seasonal planning	Monthly revenue trend	Not the main purpose
Customer spending	Total revenue by customer	Monetary value
Customer recency	Not available	Included
Repeat purchasing	Not available	Included through frequency
Customer-retention value	Limited	Stronger support
Interpretability	High	High
Reproducibility	High	High

The proposed approach will be considered successful if it produces customer groups that are clearly different in recency, frequency, and monetary value and if those groups can be connected to realistic retail actions such as retention, reactivation, loyalty recognition, or targeted engagement.

11.7 Architecture Conclusion

The system architecture provides a transparent path from raw transaction data to decision-ready analysis. Each stage produces an output required by the next stage, and all major transformations are documented in the GitHub notebooks.

This architecture supports reproducibility, reduces the risk of invalid calculations, and ensures that the final business conclusions can be traced to documented data-quality checks, feature-engineering rules, and evaluation criteria.

12. GitHub Reproducibility

A GitHub repository was created to organize the project files, document analytical progress, and allow another reviewer to understand and reproduce the workflow.

GitHub Repository: [https://github.com/Prabhjot-kaur-tech/CIND820_Retail_sales_Analytics]

12.1 Repository Structure

The repository is organized into the following folders:

Folder or File	Purpose
data/	Contains the raw Online Retail dataset and data-access documentation
notebooks/	Contains the Initial EDA, Data Cleaning and Preparation, and Business EDA notebooks
reports/	Contains the Milestone 1 report and will contain the Milestone 2 Interim Report
images/	Stores analytical figures and the pipeline diagram
README.md	Summarizes the project, dataset, analytical approach, folder structure, and execution instructions

12.2 Notebook Execution Order

The notebooks should be reviewed or executed in the following order:

1. **01_Initial_EDA.ipynb**
Reviews dataset structure, missing values, duplicate records, cancelled transactions, invalid quantities and prices, and initial revenue characteristics.
2. **02_Data_Cleaning_&_Preperation.ipynb**
Removes duplicate and invalid records, creates the cleaned sales dataset, and prepares the customer-level dataset.
3. **03_Business_EDA_&_Analysis.ipynb**
Produces product, country, monthly, distribution, and bivariate analysis outputs.

This sequence is important because the later notebooks depend on the data-quality findings and preparation decisions documented in the earlier notebooks.

12.3 Data Access and Reproduction

The raw dataset is stored in the data/ folder. The cleaned CSV files were not uploaded because they exceeded GitHub's web-upload size limit.

However, both cleaned datasets can be recreated by running the Data Cleaning and Preparation notebook. This notebook generates:

- online_retail_cleaned.csv
- online_retail_customer_cleaned.csv

This approach preserves reproducibility because the transformation steps are documented and can be executed directly from the raw dataset.

12.4 Software and Libraries

The analysis was completed in Google Colab using Python. The main libraries include:

- pandas;
- NumPy;
- Matplotlib.

12.5 Version-Control Practice

GitHub commit messages were used to document major project updates, including:

- raw dataset upload;
- Milestone 1 report upload;
- Initial EDA notebook upload;
- data-cleaning notebook upload;
- Business EDA notebook upload;
- data-folder documentation updates.

This creates a visible record of project development and helps distinguish the original project proposal from later analytical improvements.

12.6 Reproducibility Limitations

The repository supports reproduction of the main workflow, but some limitations remain:

- cleaned datasets must be regenerated because of file-size restrictions;
- Google Colab file paths may need to be adjusted when notebooks are run in a different environment;
- all notebooks should be executed in the documented order;
- the user must have access to the raw Online Retail dataset.

Despite these limitations, the repository provides the code, documentation, project structure, and analytical outputs required for a third-party reviewer to follow the project workflow.

13. Generative AI Use Declaration

I used generative AI tools to support brainstorming, organize the report structure, improve wording, clarify methodological concepts, and help understand selected Python code.

I also used generative AI to assist with the visual design of the analytical pipeline diagram. However, I decided the pipeline stages, sequence of operations, analytical methods, and final interpretations myself.

I did not use generative AI to replace the execution of the analysis, create unverified results, or make final analytical decisions. I ran all Python code in Google Colab and checked all reported values, charts, and conclusions against the notebook outputs.

I reviewed AI-assisted suggestions related to methodology, privacy, data preparation, and literature against the dataset, course requirements, official documentation, and the original scholarly sources before including them in the report.

14. Planned Next Steps and Conclusion

The Milestone 2 audit confirms that the Online Retail dataset can support product, market, time-based, and customer-value analysis, provided that its data-quality and coverage limitations are handled explicitly.

The descriptive baseline provides a transparent view of historical revenue performance across products, countries, and months. RFM analysis extends this baseline by incorporating customer recency, purchase frequency, and monetary value, making it more suitable for customer-retention and targeted-marketing decisions.

The next stage of the project will focus on completing the customer-level analysis. This work will include:

1. calculating customer-level Recency, Frequency, and Monetary variables;
2. reviewing the distributions and skewness of the RFM measures;
3. applying transparent RFM scoring and segment definitions;
4. comparing the RFM results with the revenue-ranking baseline;
5. evaluating segment size, revenue contribution, stability, and business usefulness;
6. preparing final recommendations and visual reporting.

The proposed architecture separates general sales analysis from customer-level behavioural analysis, documents the major preparation decisions, protects indirect customer identifiers, and supports reproducibility through GitHub notebooks and execution instructions.

Overall, Milestone 2 establishes a transparent and technically appropriate foundation for the final project. The next phase will convert the prepared customer-level data into interpretable RFM segments and decision-ready recommendations for inventory, marketing, and customer-retention stakeholders.

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